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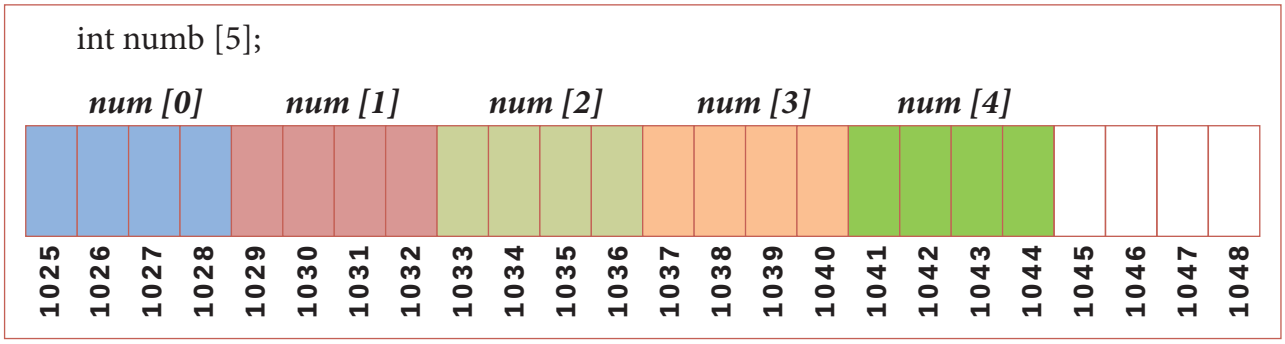

The above figure clearly shows that, the array num is an integer array with 5 elements. As per the Dev-C++ compiler, 4 bytes are allocated for every int type variable. Here, there are totally 5 elements in the array, where for each element, 4 bytes will be allocated. Totally, 20 bytes will be allocated for this array.

Datatype	Turbo C++	Dev C++
char	1	1
int	2	4
float	4	4
long	4	4
double	8	8
long double	10	10

The memory space allocated for an array can be calculated using the following formula:

Number of bytes allocated for type of array × Number of elements

Initialization

An array can be initialized at the time of its declaration. Unless an array is initialized, all the array elements contain garbage values.

Syntax:

<datatype> <array_name> [size] = {value-1,value-2,…… ,value-n};

Example

int age[5]={19,21,16,1,50};

In the above example, the array name is 'age' whose size is 5. In this case, the first element 19 is stored in age[0], the second element 21 is stored in age[1] and so on as shown in figure 12.1